DVWA on Windows Using Docker Desktop

Practical / Exam Quick Guide

Important: DVWA is intentionally vulnerable. Run it only locally on your own/authorized lab machine. Do not expose it to the LAN or public Internet.

1. Install Docker Desktop

• Install Docker Desktop for Windows.

• Start Docker Desktop and wait until Docker Engine is running.

• Open PowerShell.

2. Verify Docker

```
docker version
docker compose version
```

3. Download DVWA

Option A — using Git:

```
cd $HOME\Downloads
git clone https://github.com/digininja/DVWA.git
cd $HOME\Downloads\DVWA
```

Option B — download the DVWA ZIP, extract it, and open PowerShell in the extracted DVWA folder.

4. Confirm the Compose File

```
dir
```

You should see the Docker Compose configuration file (typically **compose.yml**).

5. Start DVWA

```
docker compose up -d
```

This starts the DVWA web container and its database container.

6. Check Containers

```
docker compose ps
```

The DVWA service should be running and the web port should be mapped to **127.0.0.1:4280**.

7. Open DVWA

Open a browser and go to: **http://localhost:4280**

If the setup page appears, click **Create / Reset Database**. If the database container has just started, wait briefly and refresh if necessary.

8. Login

Username	admin
Password	password

9. Set DVWA Security to Low

• Log in to DVWA.

• Open **DVWA Security**.

• Select **Low**.

• Click **Submit**.

10. SQL Injection Practical

Open **SQL Injection** and use the manual's local-lab test payloads:

```
1' OR '1'='1'#
```

```
'UNION SELECT user, password FROM users; #
```

Use these only against the local DVWA instance created for the practical.

11. Stored XSS Practical

Open **XSS (Stored)** and use a harmless local test such as:

```
<script>alert("XSS")</script>
```

Stored XSS is saved by the application and can execute when the stored content is displayed.

12. Reflected XSS Practical

Open **XSS (Reflected)** and use the same harmless local test:

```
<script>alert("XSS")</script>
```

Reflected XSS is immediately returned/reflected by the application rather than being stored.

13. Stop DVWA After the Practical

```
docker compose down
```

This stops and removes the containers created by the Compose project. You can start the lab again later with **docker compose up -d** from the DVWA directory.

14. Troubleshooting

Problem / Task	Command
Check running services	`docker compose ps`
View container logs	`docker compose logs`
Restart the lab	`docker compose down` `docker compose up -d`

15. Exam Quick Memory

Item	Remember
Docker check	`docker version / docker compose version`
Start DVWA	`docker compose up -d`
Check containers	`docker compose ps`
DVWA URL	`http://localhost:4280`
Login	`admin / password`
Security	`Low → Submit`
Stop	`docker compose down`

Manual compatibility note: The college manual documents XAMPP for the Windows DVWA setup. Docker is an alternative implementation that runs the same DVWA lab locally. If your examiner requires the exact installation procedure from the manual, follow the XAMPP method instead.